

better resolved.

17 A

Using $d \sin \theta = n\lambda$,

Where there is an overlap between a lower order - longer wavelength and higher order-shorter wavelength, θ is the same. Since the same diffraction grating is used, d is the same.

Hence, $n \lambda_{\text{longer}} = (n + 1) \lambda_{\text{shorter}}$

Systematically working out,

When $n = 1$ and $\lambda_{\text{longer}} = 700 \text{ nm}$, $(n+1) = 2$ and $\lambda_{\text{shorter}} = 350 \text{ nm}$. Since 350 nm is outside the range of 400 to 700 nm , there is no overlap between the first order's 700 nm and the second order's 400 nm .

When $n = 2$ and $\lambda_{\text{longer}} = 700 \text{ nm}$, $(n+1) = 3$ and $\lambda_{\text{shorter}} = 467 \text{ nm}$. Since 467 nm is within the range of 400 to 700 nm , there is an overlap between the second and third orders.

The highest order of diffraction where there is no overlap is $n = 1$, the 1st order.

Alternatively

Using $d \sin \theta = n\lambda$, work out the angle of deviation θ for 400 nm light and 700 nm for each order n .

Order, n	θ for 400 nm	θ for 700 nm
1	6.9°	12.1°
2	13.9°	24.8°
3	21.1° (inside 2 nd order)	

18 C

Statements A, B, D are correct assumptions while statement C is